

LIFELINK: INTELLIGENT BLOOD & ORGAN COORDINATION NETWORK

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1. INTRODUCTION

Hospitals depend on the timely availability of compatible blood units and viable organs to treat patients during scheduled procedures and emergencies. Blood banks and organ procurement or transplant centres largely maintain their inventories as isolated records inside individual institutions. When a hospital needs a specific blood group or a transplantable organ, staff must telephone, e-mail or visit several institutions before a match is located, consuming time that can affect patient outcomes and leaving administrators without an accurate, real-time picture of regional resource availability. Existing platforms such as **IRCS eBloodServices** [4] support digital blood requests, tracking and blood-availability search, while **BloodConnect** [5] focuses on connecting patients with voluntary donors and donor mobilisation.

National initiatives such as **e-RaktKosh** [1] and **NOTTO's e-Pratyaropan portal** [2] show that centralised, technology-driven coordination can meaningfully reduce such delays at scale. e-RaktKosh provides centralized blood-bank management and blood-availability services, while e-Pratyaropan provides a national digital platform for organ and tissue transplantation. Building on this idea at an institutional level, the proposed project, **LifeLink**, is a web-based

coordination platform that brings hospitals, blood banks, organ centres and registered donors onto a single system, replacing manual contact with structured, real-time discovery and requesting.

LifeLink provides:

- Centralised inventory visibility across institutions.
- Real-time request, acceptance and rejection workflows.
- Structured emergency broadcasting to nearby institutions and donors.
- Rule-based organ compatibility filtering, kept separate from clinical decisions.
- Blood-demand forecasting and shortage-risk prediction.
- A Smart FAQ Assistant built with Retrieval-Augmented Generation (RAG) that answers platform-usage questions from approved documentation.

2. LITERATURE SURVEY

Several existing platforms address parts of the blood and organ availability problem. **e-RaktKosh [1]**, developed by C-DAC under India's National Health Mission, centralises blood-bank inventory across more than 3,800 centres, offering standardised donor and stock records, but it does not cover organs or hospital-side request workflows. **NOTTO's e-Pratyaropan [2]** portal digitises India's organ waiting lists and allocation at hospital, state and national level, but it is built around registered transplant recipients rather than general hospital search across both blood and organs. **UNOS/UNet [3]** performs real-time organ matching and offer management in the United States and is technically mature, but it is restricted to organs and to a single country's regulatory framework, limiting its relevance to smaller regional deployments. Commercial hospital blood-bank software automates a single institution's donor and inventory registers well but does not connect one bank's stock to another's, so hospitals must still contact banks individually; many smaller banks continue to rely on spreadsheets or paper registers with no online visibility at all.

Table 1. Comparative summary of related systems

System	Scope	Key Strength	Limitation Addressed by LifeLink
e-RaktKosh (C-DAC, India)	Blood only	National scale, standardised inventory across 3,800+ centres	No organ module; no hospital-side request workflow
NOTTO e-Pratyaropan Portal (India)	Organs only	Unified national waiting list, real-time allocation	Focused on registered transplant recipients, not general resource discovery
UNOS / UNet (USA)	Organs only	Real-time donor-recipient matching, mature allocation policy engine	Restricted to a single country's regulatory framework; organs only
BloodConnect	Blood / donors	Voluntary donor mobilisation and urgent donor coordination	Reference for donor outreach
Commercial hospital blood-bank software	Single institution	Automates one bank's donor and inventory registers well	No cross-institution visibility; hospitals must still contact banks individually

System	Scope	Key Strength	Limitation Addressed by LifeLink
Spreadsheet / register-based tracking	Single institution, manual	Low cost, no infrastructure needed	No online visibility, no real-time updates, error-prone
LLM-based Assistants (GPT, Gemini, Llama, etc.)	General-purpose natural language interaction	Natural-language question answering, summarisation, information retrieval and conversational assistance	Responses may be hallucinated, unverified or from unauthenticated sources; LifeLink uses RAG over authenticated and approved documentation

Across these systems, a recurring gap is the absence of one platform that unifies blood and organ discovery for a hospital, adds real-time cross-institution notification, and layers lightweight predictive analytics and a documentation-grounded FAQ assistant on top. **LifeLink** integrates the key capabilities of these existing systems into a unified platform, addressing their identified limitations while providing additional features for real-time resource discovery, coordination, predictive analytics, and intelligent assistance.

3. PROBLEM DOMAIN

Hospitals seeking blood units or organs for transplantation must currently contact multiple blood banks, organ centres or other hospitals individually, using phone calls, e-mails or visits, because inventory data is not shared across institutions. This fragmentation delays discovery of available resources, increases the manual workload of staff, and makes it difficult for health authorities to know where a shortage is developing at a given moment.

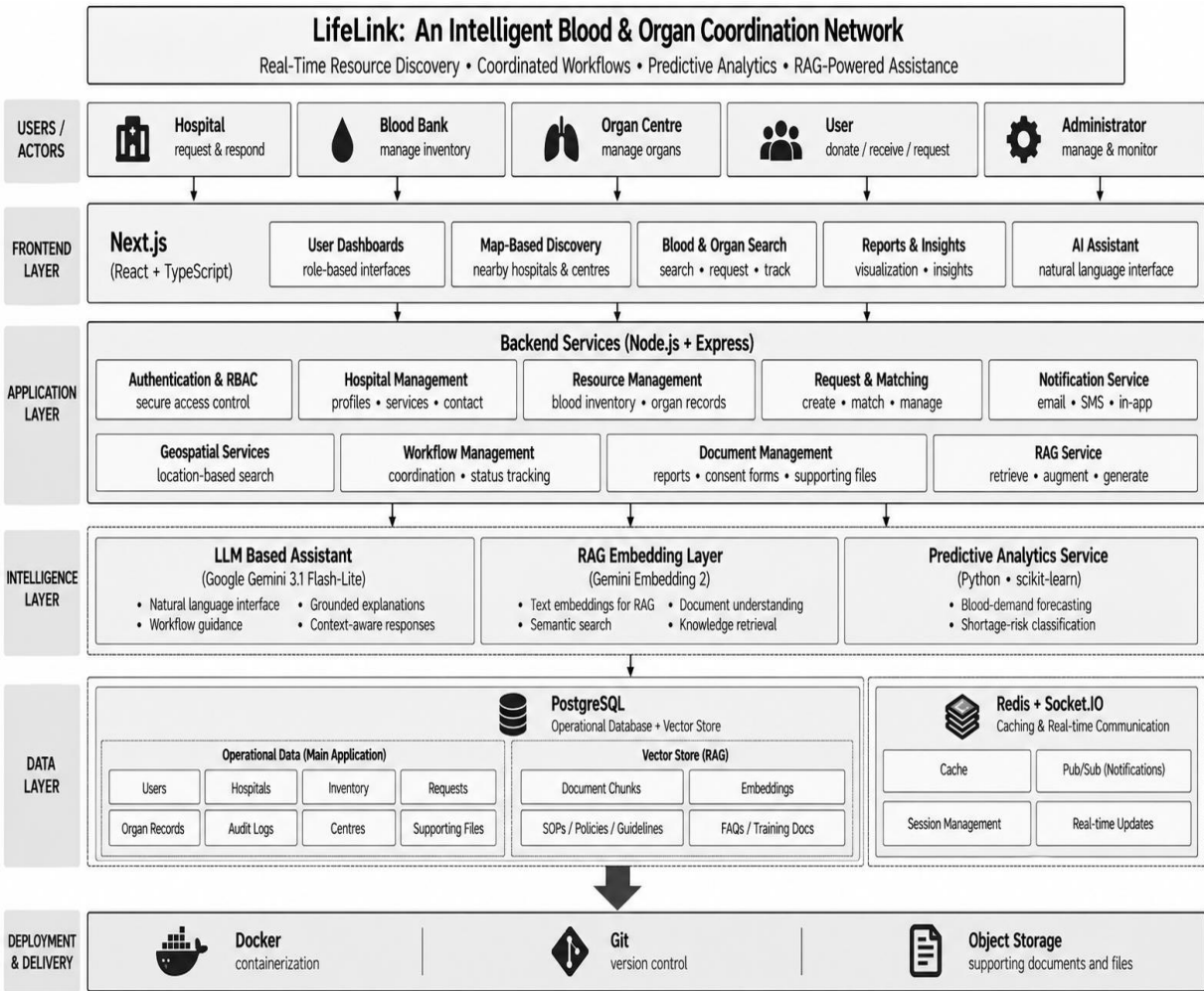
Objectives of the proposed work are to:

1. Let hospitals search live blood and organ availability across registered institutions.
2. Support structured requesting, acceptance and rejection with real-time status updates.
3. Broadcast structured emergency requests to nearby institutions and donors.
4. Maintain privacy-aware donor profiles in which contribution achievements and general availability can be shown without exposing personal contact details.
5. Apply rule-based compatibility filtering and lightweight predictive models for demand and shortage risk.
6. Provide a RAG-based FAQ assistant for platform-usage queries.
7. Maintain an auditable log of every significant action.

4. SOLUTION DOMAIN

LifeLink is a three-tier web application. The presentation tier uses Next.js, React and TypeScript with Tailwind CSS to provide role-specific dashboards for hospitals, blood banks, organ centres, donors and administrators. The application tier exposes REST APIs (Next.js server actions or Express.js) implementing authentication, role-based access control, inventory management, request workflows and notifications. The persistence tier uses PostgreSQL through the Prisma ORM, with a relational schema built around User, Hospital, BloodBank, OrganCentre, Donor, BloodUnit, Organ, BloodRequest, OrganRequest, EmergencyRequest,

Notification, Document, AuditLog and Prediction entities, related through foreign keys and indexed on fields such as blood group and location.



Blood and organ search use structured filters — blood group, quantity, location and radius — ranked by distance, expiry proximity and stock level, with nearby-institution discovery shown on a Leaflet/OpenStreetMap map. Organ matches pass through a rule-based compatibility engine presented as system-level filtering support, not a clinical decision. Requests, acceptances and rejections update inventory transactionally and propagate to dashboards over WebSockets (Socket.IO). Redis handles caching of frequent inventory queries, publish/subscribe messaging behind the real-time layer, and rate-limiting for sensitive endpoints such as emergency broadcasts.

A Python/scikit-learn service supplies regression-based blood-demand forecasting and shortage-risk classification from historical inventory and request data. A Retrieval-Augmented Generation (RAG) module retrieves relevant passages from a curated store of platform documentation and policy text and supplies them to a language model, grounding the Smart FAQ Assistant's answers to usage and policy questions in approved reference material rather than open-ended interpretation. Supporting documents uploaded by hospitals are kept in object storage, with only metadata in PostgreSQL, and every state-changing action is written to an audit log.

5. SYSTEM DOMAIN

- **Frontend:** Next.js, React, TypeScript, Tailwind CSS, for full-stack TypeScript development and responsive dashboards.
- **Backend:** Node.js (Next.js API routes/Express.js), for REST APIs and business logic in one codebase.
- **Database:** PostgreSQL with Prisma ORM, since inventories, requests and institutions form a naturally relational domain.
- **Real-time:** WebSockets/Socket.IO for instant dashboard updates; Redis for caching, pub/sub and rate-limiting.
- **Maps:** Leaflet with OpenStreetMap for location-based resource discovery and mapping.
- **Predictive analytics:** Python and scikit-learn for the demand-forecasting and shortage-risk models.
- **RAG + LLM:** Vector Database, Gemini Embedding 2 and Gemini 3.1 Flash-Lite – Semantic retrieval and grounded language assistance
- **Storage/DevOps:** Supabase for documents; Docker & Git.

6. APPLICATION DOMAIN

LifeLink applies to hospitals, blood banks and organ procurement/transplant centres operating within a city or region, and to the registered donors and administrators supporting them. It suits institutions that currently coordinate blood and organ availability manually and would benefit from a lightweight, unified alternative to national-scale platforms. Variants could support disaster-response blood mobilisation or smaller nursing homes lacking any digital inventory system. For end users, the main real-life impact is less time and effort spent locating a compatible blood unit or organ in routine and emergency situations.

7. EXPECTED OUTCOME

- A working multi-role platform (hospital, blood bank, organ centre, donor, administrator) with authentication and RBAC.
- Real-time blood and organ inventory search and discovery across registered institutions.
- End-to-end request workflows with real-time status notifications, and a structured emergency broadcast mechanism.
- Cross-institution discovery of registered blood resources and organ-related coordination records with location-aware views.
- A modular organ-coordination structure covering donor/consent, records, recipient requirements, matching/offers, procurement, transport, post-transplant and compliance.
- Working blood-demand-forecasting and shortage-risk-prediction models evaluated on synthetic/demo data.
- A RAG-based LLM assistant providing grounded, contextual answers and workflow guidance from approved documentation. Reduces repetitive platform-usage queries while enabling natural-language assistance to users and administrators.
- Administrator analytics dashboards and an auditable log of significant system actions.

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SIGNATURES

This synopsis is submitted for approval as a minor project of the Department of Computer Science & Engineering, IET DAVV, Indore.

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